

Designed by Remus Teodorescu 2026

LECTURE 12.1

Power Grid Stability Estimation

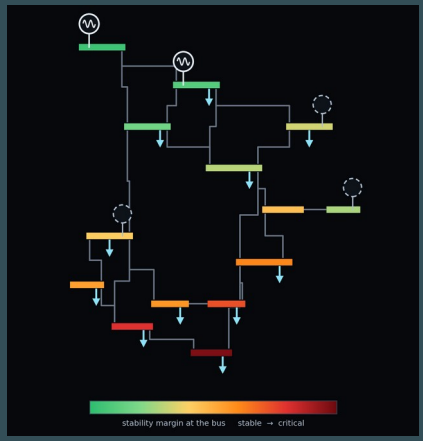
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Deep Learning for Engineering · 2026

GRID · STATE · INERTIA



A grid is a graph with the power flow equations at every bus, measured by a few noisy meters. The voltage at every bus is estimated by weighted least squares and by a physics-informed network, with no true state; then a short circuit makes a generator's rotor oscillate, and its inertia is found from the record.

Learning Objectives

SECTIONS, IN ORDER

1. **The grid** — a graph, the power flow equations, what a room sees
2. **State estimation** — least squares, and a network with no true state
3. **The swing** — the equation of motion, and the inertia from a record

UDL carries the graph network, Liu the inverse fit. The example is Ex_12.1's six-bus grid, representative of eastern Denmark and not a model of it: the line impedances are textbook values, because no operator publishes measured ones, and results that depend on them must say so.

READ, AND WHAT EACH ANSWERS

UDL ch. 13	how is a graph represented?
Liu ch. 4	what can readings determine?
SciPy	how is an ODE shot and fitted?

Recap of L11.2 Obstacle Avoidance

1 – 2 · IDENTIFYING AN OBSTACLE

1 **How can a network decide from one camera frame whether the path is blocked, where a dark-pixel rule fails?**

The network has two outputs, free and blocked, turned into a probability by the softmax and trained with cross-entropy on frames labelled by where the obstacle stands: in the lane and near. A rule that calls anything darker than the floor blocked stops for shadows and misses grey obstacles. On the rendered frames the rule is right 85 % of the time and the network 97 %, almost never stopping for a shadow; the rule misses seven grey boxes in ten, the network four. *UDL ch. 5; Ex_11 section 5*

2 **Why is the threshold on the probability of a blocked path not set at one half, and how is it chosen?**

A missed obstacle costs a collision and a false stop costs a second, so the two errors have very different costs. Lowering the threshold trades missed obstacles for false stops, and the threshold is chosen where the expected cost, each error's rate times its cost, is smallest. On the rendered frames the network misses 15 % of the obstacles at one half and 4 % at 0.05, where it stops for 5 % of the free frames; on the car both rates are measured again. *UDL ch. 5; Ex_11 section 5*

3 – 4 · DECIDING THE ACTION

3 **How does the avoidance strategy decide what the network must tell, and why does stopping alone not reach B?**

Stopping needs only free or blocked, but an obstacle that stays keeps the car standing, so it never reaches B. Steering around needs which side is free, and shifts the lane aimed at while the path is blocked, slowed to 60 %. A potential field adds a push away from every obstacle to the pull of the aim point, so it needs where each obstacle is, from a detector or a range sensor. On the course the field reaches B in 6.3 s, steering around in 8.3 s with half a centimetre to spare. *UDL ch. 19; Ex_11 section 6*

4 **Why does the rate of the control loop limit the safe speed of a car that avoids obstacles?**

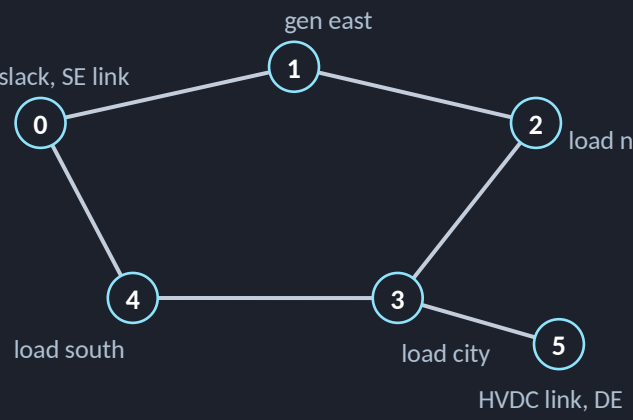
From the frame that shows the obstacle to standing still, the car drives one blind distance, its speed divided by the decisions per second, and then brakes as far as the friction of the tyres allows. That sum must stay within the range the camera judges. With a range of 0.7 m the safe speed is 1.9 m/s at 5 decisions a second and 2.7 m/s at 30; a faster network shortens only the first of the two distances. *Ex_11 section 7*

A Grid Is a Graph

Transmission grid — buses joined by lines: a graph, with a node's state being its voltage magnitude and angle, and every edge carrying an admittance. The state of the whole grid is one number pair per bus, and everything a control room wants to know is a function of it.

EX_12.1'S SIX BUSES, AS THEY ARE WIRED

SIX BUSES, SIX LINES, ONE RING



a bus is a node, numbered as in the notebook

each line is an edge, weighted by its admittance $G + jB$

a ring of five buses, and one radial line to the HVDC bus

THE CASE, AND ITS LIMITS

the grid	6 buses, 6 lines, 100 MVA base
the state	6 magnitudes, 5 angles
stands for	eastern Denmark, loosely
impedances	textbook values, flagged
so	say when a result leans on them

No operator publishes measured impedances; representative is the honest word.

The Power Flow Equations

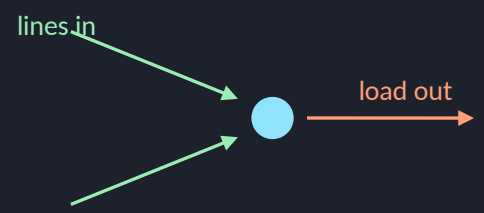
Power flow equations — two algebraic balances at every bus: the active and the reactive power the lines bring must equal what the bus injects or takes. They are constraints the voltages satisfy at every instant, not a simulation in time - the elliptic habit of L8.1, on a graph.

TWO BALANCES AT EVERY BUS

$$P_i = V_i \sum_k V_k (G_{ik}\cos\theta_{ik} + B_{ik}\sin\theta_{ik}), \qquad Q_i = V_i \sum_k V_k (G_{ik}\sin\theta_{ik} - B_{ik}\cos\theta_{ik})$$

V_i, θ_i the voltage magnitude and angle at bus i : the state, G, B the admittance matrix, $\theta_{ik} = \theta_i - \theta_k$

ONE BUS, ONE BALANCE



what the lines bring

equals what the bus takes

two balances: P and Q

THREE WAYS THE EXERCISE USES THEM

1 · solver	Newton, all injections given: the truth
2 · check	mismatch ΔP, ΔQ of any estimate, per bus
3 · loss	minimise the mismatch, in torch

Why a loss: in operation nobody knows the true state, so there is no label to compare an estimate with. The physics can still judge it, since the true state leaves zero mismatch wherever the injection is known: at a meter, a forecast or a bus with nothing connected. Written in torch, the mismatch passes its gradient back through the voltages to the weights.

What the Control Room Sees

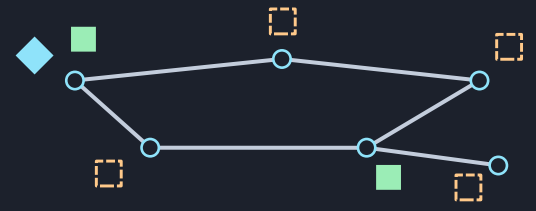
Measurement model — every reading is a function of the state plus noise: a PMU reads a bus's magnitude and angle, a meter its P and Q, a forecast stands in where no meter exists. The thin set has fewer readings than unknowns, and where they sit on the graph decides everything.

ONE LINE PER METER

$$z_j = h_j(V, \theta) + \varepsilon_j$$

z_j one reading, h_j what that meter measures as a function of the state, ε_j its noise

THE FORECAST SET ON THE GRID



PMU: $|V|$ and θ at its bus
meter: P, Q injected there
forecast stands in, loosely
shown: the forecast set, the set the network uses

TALK · 2 MIN How would you explain, in your own words, why a grid's state can be known better than its meters? Then, from the table on the right, why does the thin set fail where the forecast set does not?

THE THREE SETS OF EX_12.1

set	readings	covers
full	14 = 4 + 10	every unknown
thin	6	5 of 11 unknowns
forecast	6 + priors	every unknown, loosely

Full: PMUs at buses 0 and 3 read V, θ(4), meters at buses 0 – 4 read P, Q(10): 14 readings for 11 unknowns. Bus 5 has no meter, yet the balance at bus 3 sums over its neighbours, so P_3, Q_3 fix bus 5 too. Coverage is where the meters sit on the graph, not how good they are.

Weighted Least Squares

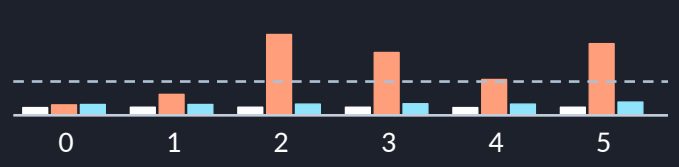
State estimation — finding the voltage magnitude and angle at every bus from fewer, noisy readings. The estimator every utility runs is weighted least squares: each meter's misfit, scaled by its noise, minimised by Gauss–Newton iterations. It is the baseline any network must face.

THE ESTIMATOR IN SERVICE

$$\hat{x} = \arg\min_{x} \sum_{j} \frac{(z_j - h_j(x))^2}{\sigma_j^2} \qquad \sigma_j^2 = \mathrm{E}\left[\varepsilon_j^2\right]$$

z_j the reading of meter j, $h_j(x)$ what meter j would read if the state were x, $\hat{x}$ the estimated state
σ_j the standard deviation of the noise ε_j of meter j, $\mathrm{E}[\,\cdot\,]$ the expected value : the mean over many readings

ERROR IN THE VOLTAGE MAGNITUDE AT EACH BUS



full set, 14 readings
thin set, 6: worst at buses 2, 5, 3
forecast set
dashed: the target, 0.01 p.u.

Why it works: each reading is an equation in V, θ through h_j. With independent normal noise the weighted sum is minus twice the log-likelihood, so $\hat{x}$ is the most probable state.

WORST BUS ERROR, MEAN OF 300 POINTS

set	in $\lvert V \rvert$	angle	worst bus
full, 14	0.0030 p.u.	0.07°	5, the HVDC
thin, 6	0.0248 p.u.	1.82°	2, load north
forecast	0.0050 p.u.	0.15°	5, the HVDC

Each value is the error at the worst bus of an operating point, averaged over 300 points. The exercise sets its target at 0.01 p.u., every bus within 1 % of its nominal voltage. One estimate takes about 2 ms on a laptop processor: eleven unknowns and a few iterations. The thin set fails because its six readings do not determine six of the eleven unknowns.

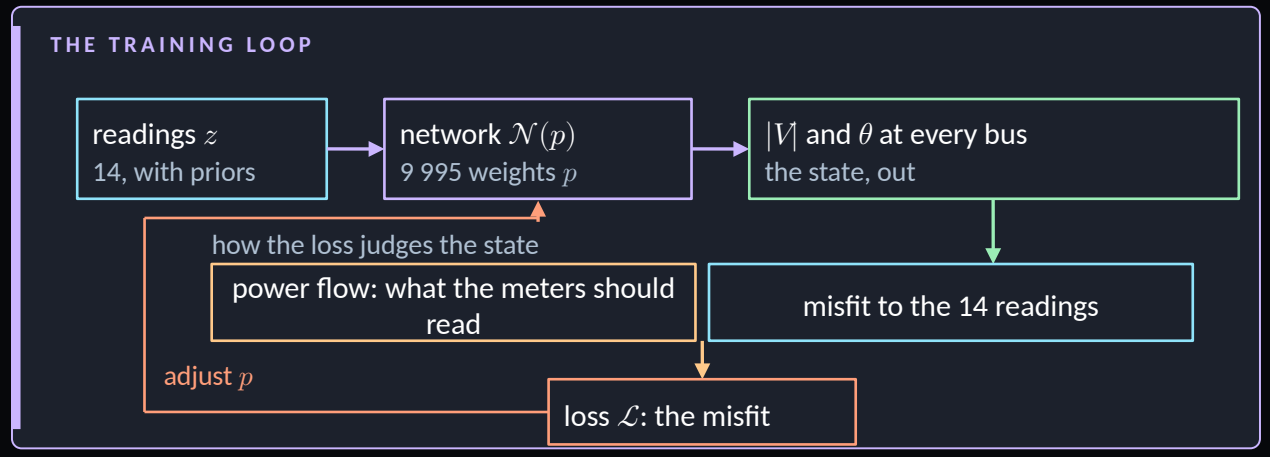
A Network Constrained by Physics

Physics-constrained estimation — a second estimator: a network maps the readings to the state at every bus. Its loss is the weighted misfit of least squares, averaged over many operating points, with the power flow equations inside h_j. No true state appears anywhere in the loss.

THE TRAINING LOSS: WEIGHTED MISFIT, NO TRUE STATE

$$\mathcal{L} = \frac{1}{N}\sum_{n}\sum_{j}\frac{\left(z_j^{(n)} - h_j(\hat{x}^{(n)})\right)^2}{\sigma_j^2}, \qquad \hat{x}^{(n)} = \mathcal{N}(z^{(n)}; p)$$

$\mathcal{N}$ the network, p its weights, n one of N operating points; h_j of a P or Q reading is the power flow equations a bus with no meter enters through a forecast of its injection, as a reading with a large σ_j; no true state in the loss



AGAINST THE BASELINE, HONESTLY

accuracy	0.0043 p.u.: it does not win
one estimate	0.26 ms; 2 µs in a batch
training	13 s, once
a line out, untold	both fail alike
but the check	its misfit J: 0.28 to 2.35

The gain is the time of one estimate, and a self-check that says when not to trust it.

The Swing Equation

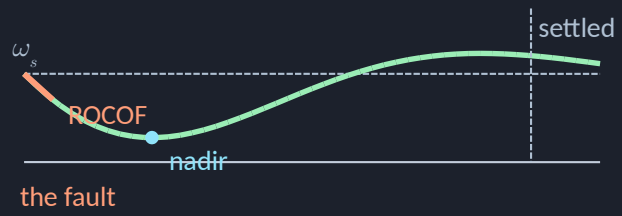
Swing equation — the grid set in motion: a machine's rotor accelerates with the gap between its mechanical and its electrical power, held back by inertia and damping, and the electrical power ties the machines together. One ordinary differential equation per machine.

THE EQUATION OF MOTION

$$\frac{d\delta}{dt} = \omega - \omega_s, \qquad \frac{2H}{\omega_s}\frac{d\omega}{dt} = P_m - P_e(\delta) - D\frac{\omega - \omega_s}{\omega_s}$$

δ the rotor angle, ω its speed, H the inertia in seconds, D damping, P_m, P_e mechanical and electrical power

AFTER A FAULT



ROCOF: how fast it first falls

nadir: the deepest deviation

settling time: until it dies away

more renewables: H falls, the dip faster and deeper

WHY INERTIA IS THE NUMBER

H	seconds of stored spin
falling	converters replace machines
so	the grid swings faster
and	operators must know today's H
from what	the disturbances it records

DK2 swings with the Nordic area, so the Nordic frequency is the right record for a DK2 plant.

The Inertia From a Record

Inertia identification — the inverse problem of the block: H and D trained beside the weights so the swing equation fits a recorded disturbance, as L8.2's diffusivity was. The residual is written with H on the left, so a large H cannot hide, and only the record after the fault is used.

THE FIT, WRITTEN SO IT CANNOT HIDE

$$H, D \text{ trainable}, \qquad \mathcal{L} = \left\| \frac{2H}{\omega_s}\dot{\omega} - P_m + P_e + D\,\Delta\omega \right\|^2 + w_D\,\mathcal{L}_{record}$$

$\mathcal{L}_{record}$ the misfit to the recorded frequency, after the fault is cleared; H on the left, so a large H cannot hide

THE FIT IN PYTORCH

```
H = torch.nn.Parameter(torch.tensor(8.0))        # trained with the weights
res = (2*H/w_s)*grad(w, t) - P_m + P_e + D*dw     # H on the left
L = res.pow(2).mean() + W_D*(w - rec).pow(2).mean()   # after the fault
```

TALK · 2 MIN How would you decide, in your own words, which part of a record identifies a parameter? Then, from the table on the right, why does the window before the fault fail from both starts?

THE INERTIA EACH FIT FINDS, TRUE 4.00 s

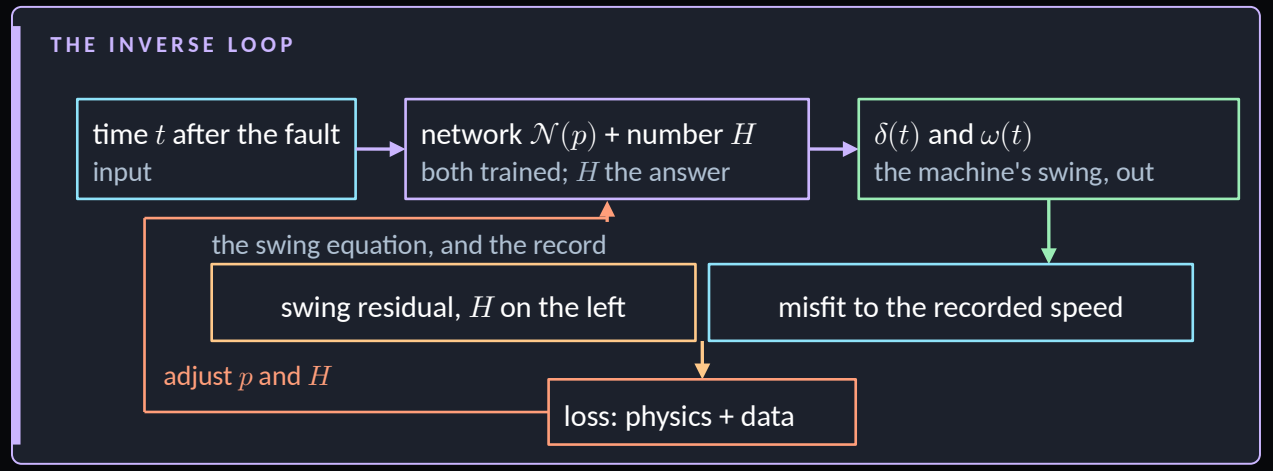
fit	start 8 s	start 2 s
shooting, after	7.41 s	1.83 s
physics-informed, after	3.89 s	3.90 s
shooting, before	50 s, the bound	1.94 s
physics-informed, before	8.35 s	2.10 s

Shooting has local minima on this record; the physics-informed fit did not - a finding about this record and these starts, not a law. Before the fault nothing excites the machine, and no method identifies what the data does not carry.

Solving the Inverse Problem

Inverse training loop — the fit of the last slide drawn as a training loop: the time since the fault goes in, the network returns the angle and the speed, judged by the swing residual, the trainable inertia inside it, and by the misfit to the recorded speed. The weights and the inertia adjust together.

THE SAME TRAINING, ONE MORE TRAINABLE NUMBER



WHAT THE INVERSE PROBLEM NEEDS

the data	the recorded speed, 0.8 s
trained	9 995 weights, and H
built in	H on the residual's left
the answer	H = 3.89 s, 3 % under
the scaffolding	the network itself

The network is scaffolding: the number carried out of the training is the inertia.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
bad-data detection	a wrong meter, found by its residual	Liu ch. 4	every estimator in service
placing one more meter	where a reading buys observability	pandapower docs	the thin set's missing half
the critical clearing time	how long a fault may stay	PowerSystems.jl	protection settings
rate of change of frequency	the first seconds of a real record	ENTSO-E data	low-inertia grids
the graph network's layers	messages passed along the lines	UDL ch. 13	grids too large for one MLP; L12.2

WHY THEY ARE LEFT OUT

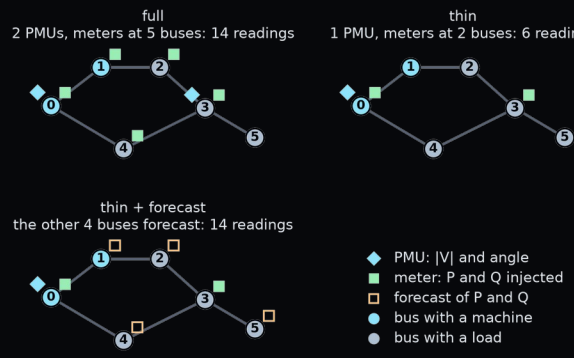
Ex_12.1 estimates one six-bus grid from fourteen readings and fits one machine's inertia, scored against Newton's power flow, so none of them is needed to do it. The last row is where the next lecture begins.

EXERCISE

Exercise 12.1

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the grid, the three sets of readings, the power flow
- **4** — Newton–Raphson reference; least squares on each set
- **5** — the physics-informed network; then a line out, untold
- **6** — the inertia from the recorded disturbance, two ways
- **7 – 9** — the report; mini projects MP12.1A and MP12.1B



THE QUESTION EACH PART ANSWERS

1 – 3	what is the state of a grid?
4	how far do the meters carry?
5	what replaces the true state?
6	what does a record identify?

REPORT QUESTION, TO CONCLUDE

The network matched least squares from the same readings and answered three hundred estimates in a blink; its residual rose eightfold when a line went out untold. In a control room, what would you use each tool for?

SUMMARY

Key Takeaways

STATE ESTIMATION	
what it is	the state from few, noisy meters
function	every bus's V and θ
train, tune	no true state: physics in the loss
vs least squares	equal accuracy, a fraction of the time
used for	a room's picture, and its self-check

SWING DYNAMICS	
what it is	the grid's equation of motion
function	angles and speeds after a fault
train, tune	H, D beside the weights
vs shooting	no local minima on this record
used for	today's inertia, from records

Next — L12.2: whether the grid survives the next fault, predicted before it happens.

Questions

1 – 2 · A NETWORK FOR THE STATE

1 **How can a network estimate the state of the grid from the readings when no true state is available for training?**

The network takes the fourteen readings of the forecast set and returns the voltage magnitude and angle at every bus. From the state it proposes, the power flow equations give what every meter should read, and the loss is the weighted misfit between those values and the actual readings. The reference angle is fixed at zero and the magnitudes are held between 0.90 and 1.10 per unit by construction. No true state appears anywhere in the training. *UDL ch. 13; Ex_12.1 section 5*

2 **When does a network for the state of the grid pay against weighted least squares, and how is its estimate checked?**

From the same readings it reaches the same accuracy as weighted least squares, so it does not gain in accuracy. Its gain is time: once trained, one estimate takes a quarter of a millisecond, and a few microseconds each in a batch. Both rest on the model of the grid, and with a line out and nobody told both are wrong. The misfit of the network's state to the readings then rises about eightfold, a check that needs no true state. *Liu ch. 4; Ex_12.1 sections 4 – 5*

3 – 4 · THE INERTIA FROM A RECORD

3 **How can the inertia of a machine be estimated with a network from a recorded disturbance?**

A network takes the time after the fault and returns the rotor angle and speed of the machine, and the inertia is one more trainable number, kept positive as an exponential. The loss is the residual of the swing equation, computed by autograd, plus the misfit to the recorded frequency. The residual is written with the inertia multiplying the acceleration: divided by it, a very large inertia makes the residual small whatever the swing does. *Liu ch. 4; Ex_12.1 section 6*

4 **Which part of a record identifies the inertia, and how does the network fit compare with the classical fit?**

Only the window after the fault, where the machine swings, carries the inertia. Before the fault nothing moves, and every fit returns a number near its own start or at its bound, with a perfect misfit. After the fault the network fit returned 3.89 s from both starts against the true 4.00 s, while the classical fit by integration stayed near where it was started. That is a finding on this record, and a fit is believed only when two starts agree. *Liu ch. 4; Ex_12.1 section 6*

NEXT

L12.2 — Power Grid Stability Prediction

From what the grid is to what it will do: whether it survives the next fault, predicted by a graph network in the time a protection relay has.

BEFORE THE NEXT SESSION

read	UDL ch. 13
run	Ex_12.1, sections 1 to 6
bring	your inertia, from both starts
think about	what the residual map told you

The estimation half of the block is done: the state, and the machine behind it.